
Hong LIU

Basic Information

Gender: Male Date of birth: 1990-02-07

Address: 422 Siming South Road, Siming District, Xiamen, Fujian, 361005, China

Phone: (86) 18359707753

E-mail: hlynn@xmu.edu.cn or lynnlou.xmu@gmail.com or hliu@ids.osaka-u.ac.jp

Website: <https://lynnhongliu.github.io/hliu/>

EMPLOYMENT

Associate Professor	2025.04 - Now
School of Informatics, Xiamen University, China	
Specially-Appointed Assistant Professor	2024.04 - 2025.03
Institute for Datability Science, Osaka University, Japan	
Specially-appointed Researcher	2024.01 - 2024.03
Institute for Datability Science, Osaka University, Japan	
Postdoctoral Researcher	2020.10 - 2023.12
Satoh-Lab, National Institute of Informatics, Japan	

EDUCATION

Xiamen University, Ph.D, Computer Science, China, 2020

Doctoral Thesis Title: Towards Compact Viusal Features

Jiangxi Normal University, M.S, Computer Science, China, 2015

Hubei University of Automotive Technology, B.S, Electronic Information Engineering, China, 2012

RESEARCH INTERESTS

Computer Vision and Machine Learning

- AI safty & reliability & ethics
- Visual representation learning
- Large scale image/text/audio/video retrieval

Peer-reviewed Journal:

- Sisuo Lyu, **Hong Liu** (*Corresponding Author*), Jie Li, Yan Teng, Yingchun Wang.
Improving Foundation Model Group Robustness with Auxiliary Sentence Embeddings.
Transactions on Machine Learning Research, 2026.
- Xiaotian Si, Linghui Li, Zhihao Tang, Bingyu Li, Liwei Zhang, Kaiguo Yuan, **Hong Liu**, Qi Tian.
Quality-Agnostic Deepfake Detection with Saliency-Guided Restoration and Adaptive Fusion.
IEEE Internet of Things Journal, 2026.
- Ke Sun, Zhongxi Chen, Xianming Lin, Xiaoshuai Sun, **Hong Liu** (*Corresponding Author*), Rongrong Ji.
Conditional Diffusion Models for Camouflaged and Salient Object Detection.
IEEE Transactions on Pattern Analysis and Machine Intelligence, 2025.

-
- Zhun Zhong, **Hong Liu** (*Corresponding Author*), Yin Cui, Shin'ichi Satoh, Nicu Sebe, Ming-Hsuan Yang. *Guest Editorial: Special Issue on Open-World Visual Recognition*. International Journal of Computer Vision, 2025.
 - Hao Zhang, Wenqi Shao, **Hong Liu**, Yongqiang Ma, Ping Luo, Yu Qiao, Kaipeng Zhang *Towards Evaluating the Robustness of Large Vision-Language Model on Adversarial Visual-Instructions* IEEE Transactions on Information Forensics and Security, 2024.
 - Huafeng Kuang, **Hong Liu** (*Corresponding Author*), Xianming Lin, Rongrong Ji. *Defense Against Adversarial Attacks Using Topology Aligning Adversarial Training*. IEEE Transactions on Information Forensics and Security, 2024.
 - Zelong Zeng, Fan Yang, **Hong Liu** (*Corresponding Author*), Shin'ichi Satoh. *Improving deep metric learning via self-distillation and online batch diffusion process*. Visual Intelligence, 2024.
 - **Hong Liu**, Yongqing Sun, Yukihiro Bandoh, Masaki Kitahara, Shin'ichi Satoh. *Deep Counterfactual Representation Learning for Visual Recognition against Weather Corruptions*. IEEE Transactions on Multimedia, 2023.
 - **Hong Liu**, Zhun Zhong, Nicu Sebe, Shin'ichi Satoh. *Mitigating Robust Overfitting via Self-Residual-Calibration Regularization*. Artificial Intelligence Journal, 2023. (Also with IJCAI Journal Track, 2024)
 - Huafeng Kuang, **Hong Liu**, Yongjian Wu, Rongrong Ji. *Semantically Consistent Visual Representation for Adversarial Robustness*. IEEE Transactions on Information Forensics and Security, 2023.
 - Zhengwei Yang, Xian Zhong, Zhun Zhong, **Hong Liu**, Zheng Wang, Shin'ichi Satoh. *Win-Win by Competition: Auxiliary-Free Cloth-Changing Person Re-Identification*. IEEE Transaction on Image Processing, 2023.
 - Fengxiang Yang, Juanjuan Weng, Zhun Zhong, **Hong Liu**, Zheng Wang, Zhiming Luo, Donglin Cao, Shaozi Li, Shin'ichi Satoh, and Nicu Sebe. *Towards Robust Person Re-identification by Defending Against Universal Attackers*. IEEE Transactions on Pattern Analysis and Machine Intelligence. 2022.
 - Deng-Ping Fan, Ziling Huang, Peng Zheng, **Hong Liu** (*Corresponding Author*), Xuebin Qin, Luc Van Gool. *Deep Facial Synthesis: A New Challenge*. Machine Intelligence Research, 2022.
 - Mingbao Lin, Rongrong Ji, **Hong Liu**, Xiaoshuai Sun, Shen Chen, Qi Tian. *Hadamard Matrix Guided Online Hashing*. International Journal of Computer Vision, 2020.
 - **Hong Liu**, Rongrong Ji, Jingdong Wang, Chunhua Shen. *Ordinal Constraint Binary Coding for Approximate Nearest Neighbor Search*. IEEE Transaction on Pattern Analysis and Machine Intelligence, 2019.
 - Rongrong Ji, **Hong Liu** (*Corresponding Author*), Liujuan Cao, Di Liu, Yongjian Wu, Feiyue Huang. *Towards Optimal Manifold Hashing via Discrete Locally Linear Embedding*. IEEE Transaction on Image Processing, 2017.
 - **Hong Liu**, Aiwen Jiang, Mingwen Wang, Jianyi Wan. *Local Similarity Preserved Hashing Learning via Markov Graph for Efficient Similarity Search*. Neurocomputing, Volume 159, pp.144-150, 2015.

Peer-reviewed Conference:

- Yaoyu Jin, Xiaochun Yang, **Hong Liu** (*Corresponding Author*), Leixia Wang, Jian Li, Rui Ding, Bin Wang. Towards Copyrighted Characters Infringement Detection via Copyright-Aware Prompt Learning. IEEE/CVF Computer Vision and Pattern Recognition Conference, 2026.
- Liyun Zhang, Zheng Lian, Hong Liu, Takanori Takebe, Shozo Nishii, Yuta Nakashima. QuMAB: Query-based Multi-annotator Behavior Pattern Learning. AAAI Conference on Artificial Intelligence, 2026. (Oral)
- Liyun Zhang, Zheng Lian, **Hong Liu** (*Corresponding Author*), Takanori Takebe, Yuta Nakashima. SimLabel: Similarity-Weighted Semi-supervision for Multi-annotator Learning with Missing Labels. AAAI Conference on Artificial Intelligence, 2026.
- **Hong Liu**, Yuta Nakashima, Noboru Babaguchi. *Paladin: Understanding Video Intentions in Political Advertisement Videos*. IEEE/CVF Winter Conference on Applications of Computer Vision, 2025.
- Yongqing Sun, **Hong Liu**, Qiong Chang, Xian-hua Han. *Deep Dual Internal Learning for Hyperspectral Image Super-Resolution*. International Conference on Multimedia Modelling, 2025.
- Ke Sun, Shen Chen, Taiping Yao, **Hong Liu** (*Corresponding Author*), Xiaoshuai Sun, Shouhong Ding, Rongrong Ji. *DiffusionFake: Enhancing Generalization in Deepfake Detection via Guided Stable Diffusion*. Neural Information Processing Systems, 2024.
- Yue Yang, Yuqi Lin, **Hong Liu**, Wenqi Shao, Runjian Chen, Hailong Shang, Yu Wang, Yu Qiao, Kaipeng Zhang, Ping Luo. *Towards Implicit Prompt For Text-To-Image Models*. International Conference on Machine Learning, 2024.
- Antoine Gratia, **Hong Liu**, Shin'ichi Satoh, Paul Temple, Pierre-Yves Schobbens, Gilles Perrouin. *CNNGen: A Generator and a Dataset for Energy-Aware Neural Architecture Search*. European Symposium on Artificial Neural Networks. 2024.
- Huafeng Kuang, **Hong Liu**#, Yongjian Wu, Shin'ichi Satoh, Rongrong Ji. *Improving Adversarial Robustness via Information Bottleneck Distillation* (# Project lead). Neural Information Processing Systems, 2023.
- Zhenglin Zhou#, Huaxia Li#, **Hong Liu**#, Nanyang Wang, Gang Yu, Rongrong Ji. (# contribute equally) *STAR Loss: Reducing Semantic Ambiguity in Facial Landmark Detection*. IEEE/CVF Computer Vision and Pattern Recognition Conference, pp.15475-15484, 2023.
- Ke Sun, **Hong Liu**, Taiping Yao, Xiaoshuai Sun, Shen Chen, Shouhong Ding, Rongrong Ji. *An Information Theoretic Approach for Attention-Driven Face Forgery Detection*. European Conference on Computer Vision, pp.111-127, 2022.
- Yixu Wang, Jie Li, **Hong Liu**, Yongjian Wu, Rongrong Ji. *Black-Box Dissector: Towards Erasing-based Hard-Label Model Stealing Attack*. European Conference on Computer Vision, pp.192-208, 2022.
- Zhengwei Yang, Xian Zhong, **Hong Liu**, Zhun Zhong, Zheng Wang. *Attentive Decoupling Network for Cloth-Changing Re-identification*. IEEE International Conference on Multimedia and Expo, 2022.
- Nobukatsu Kajiura, **Hong Liu**, Shin'ichi Satoh. *Improving Camouflaged Object Detection with the Uncertainty of Pseudo-edge Labels*. ACM International Conference on Multimedia in Asia, pp.1-7, 2021.
- Xinshuai dong, Anh Tuan Luu, Rongrong Ji, **Hong Liu**. *Towards Robustness Against Natural Language Word Substitutions*. (Spotlight Paper) International Conference on Learning Representations, pp.1-14, 2021.

-
- Ke Sun, **Hong Liu**, Qixiang Ye, Yue Gao, Jianzhuang Liu, Ling Shao, Rongrong Ji.
Domain General Face Forgery Detection by Learning to Weight.
AAAI Conference on Artificial Intelligence, pp. 2638-2646, 2021.
 - Fengxiang Yang, Zhun Zhong, **Hong Liu**, Zheng Wang, Zhiming Luo, Shaozi Li, Nicu Sebe, Shin'ichi Satoh.
Learning to Attack Real-World Models for Person Re-identification via Virtual-Guided Meta-Learning.
AAAI Conference on Artificial Intelligence, pp. 3128-3135, 2021.
 - Xinshuai Dong, **Hong Liu**, Liujuan Cao, Rongrong Ji, Qixiang Ye, Jianzhuang Liu, Qi Tian.
API-Net: Robust Generative Classifier via a Single Discriminator.
European Conference on Computer Vision, pp. 379-394, 2020.
 - Hanlin Chen, Baochang Zhang, Song Xue, Xuan Gong, **Hong Liu**, Rongrong Ji, David Doermann.
Anti-Bandit Neural Architecture Search for Model Defense.
European Conference on Computer Vision, pp.70-85, 2020.
 - Jie Li, Rongrong Ji, **Hong Liu**, Jianzhuang Liu, Bineng Zhong, Cheng Deng, Qi Tian.
Projection & Probability-Driven Black-Box Attack.
IEEE/CVF Computer Vision and Pattern Recognition Conference, pp.362-371, 2020.
 - **Hong Liu**, Rongrong Ji, Jie Li, Baochang Zhang, Yue Gao, Yongjian Wu, Feiyue Huang.
Universal Adversarial Perturbation via Prior Driven Uncertainty Approximation. (Oral Paper)
IEEE/CVF International Conference on Computer Vision, pp.2941-2949, 2019.
 - **Hong Liu**, Jie Li, Rongrong Ji, Yongjian Wu.
Learning Neural Bag-of-Matrix-Summarization with Riemannian Network.
AAAI Conference on Artificial Intelligence, pp.8746-8753, 2019
 - Jie Li, Rongrong Ji, **Hong Liu**, Xiaopeng Hong, Yue Gao, Qi Tian.
Universal Perturbation Attack Against Image Retrieval.
IEEE/CVF International Conference on Computer Vision, pp.4899-4908, 2019.
 - Huafeng Kuang, Rongrong Ji, **Hong Liu**, Shengchuan Zhang, Xiaoshuai Sun, Feiyue Huang, Baochang Zhang.
Multi-modal Multi-layer Fusion Network with Average Binary Center Loss for Face Anti-spoofing.
ACM International Conference on Multimedia 2019, pp.48-56, 2019.
 - Jie Hu, Rongrong Ji, **Hong Liu**, Shengchuan Zhang, Cheng Deng, Qi Tian.
Towards Visual Feature Translation.
IEEE / CVF Computer Vision and Pattern Recognition Conference, pp. 2999-3008, 2019.
 - Mingbao Lin, Rongrong Ji, **Hong Liu**, Xiaoshuai Sun, Yongjian Wu, Yunsheng Wu.
Towards Optimal Discrete Online Hashing with Balanced Similarity.
AAAI Conference on Artificial Intelligence, pp. 8722-8729, 2019.
 - **Hong Liu**, Mingbao Lin, Shengchuan Zhang, Yongjian Wu, Feiyue Huang, Rongrong Ji.
Dense Auto-Encoder Hashing for Robust Cross-Modality Retrieval.
ACM International Conference on Multimedia 2018, pp.1589-1597, 2018.
 - Mingbao Lin, Rongrong Ji, **Hong Liu**, Yongjian Wu.
Supervised Online Hashing via Hadamard Codebook Learning. (Oral Paper)
ACM International Conference on Multimedia 2018, pp.1635-1643, 2018.
 - Jianqiang Qian, Xianmin Lin, **Hong Liu**, Youming Deng, Rongrong Ji.
Towards Compact Visual Descriptor via Deep Fisher Network with Binary Embedding. (Oral Paper)
International Conference on Multimedia and Expo, pp.1-6, 2018.
 - **Hong Liu**, Rongrong Ji, Yongjian Wu, Feiyue Huang, Baochang Zhang.
Cross-Modality Binary Code Learning via Fusion Similarity Hashing.
IEEE/CVF Computer Vision and Pattern Recognition Conference, pp.7380-7388, 2017.

-
- **Hong Liu**, Rongrong Ji, Yongjian Wu, Feiyue Huang.
Ordinal Constrained Binary Code Learning for Nearest Neighbor Search. (Oral Paper)
AAAI Conference on Artificial Intelligence, pp.2238-2244, 2017.
 - **Hong Liu**, Rongrong Ji, Yongjian Wu, Gang Hua.
Supervised Matrix Factorization for Cross-Modality Hashing. (Oral Paper)
International Joint Conference on Artificial Intelligence, pp. 1767-1773, 2016.
 - **Hong Liu**, Rongrong Ji, Yongjian Wu, Wei Liu.
Towards Optimal Binary Code Learning via Ordinal Embedding.
AAAI Conference on Artificial Intelligence, pp.1258-1265, 2016.

Inventions and Patents:

- Rongrong Ji, Xianming Lin, **Hong Liu**, *A kind of ultralow complexity image search method retaining Hash based on sequence*, China Application, CN106777038B
- Sun Yongqing, Bando Yukihiro, Hiwazaki Yuusuke, **Hong Liu**, Shin'ichi Satoh. *Image Object Detection Device, Detection Method for Image Object and Program*, Japanese Application, P2023-167528A.

Non-Peer-Reviewed Paper:

- **Hong Liu**, Yongqing Sun, Shin'ichi Satoh.
Rethinking Robust 3D Recognition via Multi-view Test-Time Adaptation.
26th Meeting on Image Recognition and Understanding (MIRU), 2023.
- **Hong Liu** and Shin'ichi Satoh.
Rethinking Adversarial Training with A Simple Baseline.
arXiv:2306.07613, 2023.
- **Hong Liu**.
Sparse-Inductive Generative Adversarial Hashing for Nearest Neighbor Search.
arXiv:2306.06928, 2023.
- Xingbin Liu, Huafeng Kuang, **Hong Liu**, Xianming Lin, Yongjian Wu, Rongrong Ji.
Latent Feature Relation Consistency for Adversarial Robustness.
arXiv:2303.16697, 2023.
- Xiao Liu, Shengchuan Zhang, **Hong Liu**, Xin Liu, Cheng Deng, Rongrong Ji.
CerfGAN: A Compact, Effective, Robust, and Fast Model for Unsupervised Multi-Domain Image-to-Image Translation.
arXiv:1805.10871, 2018

PROJECTS AND GRANTS

Social Information Technologies to Counter Infodemics, JST CREST Project (Join in 2024)

Provable, Explainable, Efficient, Robust Deep Neural Network for Large-scale Multimedia Retrieval, JSPS International PostDoc Project (2021-2023)

NII-NTT Collaborative Project (2021-2023)

PROFESSIONAL EXPERIENCE

Guest Editor of IJCV Special Issue on Ensuring Trustworthiness in Open-World Visual Recognition

Guest Editor of IJCV Special Issue on Open-World Visual Recognition.

Area Chair of ACM Conference on Multimedia 2024, 2025, 2026

Workshop Organizer on CVPR 2025, ACM MM 2025

Special Session on MMM 2026

Associate Editor of Visual Intelligence

Guest Editor of Electronics Special Issue on Adversarial Machine Learning: Attacks, Defenses and Security.

Guest Editor of Electronics Special Issue on Multimedia Content Analysis, Management and Retrieval: Trends and Challenges.

Journal Reviewer: IEEE TPAMI, IJCV, IEEE TIP, ACM TOPS, IEEE TMM, IEEE TKDE, IEEE TBD, IEEE TNNLS, IEEE TC, ACM TIST, IEEE TAI, PR, PRL, AIRE, KBS, NEUCOM, TVCJ, PLOS ONE, SIGNAL PROCESS-IMAGE.

Conference reviewer or PC member: ICLR, ICML, NeurIPS, CVPR, ICCV, ECCV, WACV, IJCAI, AAAI, ICASSP, ACM MM, ACCV, ICMR.

AWARDS

Fujian Provincial Patent Award, 2024

Notable Reviewer, ICLR 2023

Top-100 Chinese New Stars in Artificial Intelligence by Baidu Scholar, China, 2021

Japan Society for the Promotion of Science (JSPS) Fellowship, Japan, 2020

Outstanding Doctoral Dissertation Award, China Society of Image and Graphics (CSIG), China, 2020

Fujian Outstanding Doctoral Dissertation Award, Fujian, China, 2020

Outstanding Ph.D. Graduate Student, Xiamen University, China, 2020

INVITED TALKS

Implicit Prompt For Text-To-Image Models, CCIG, Xi'an, 2024.

Understanding Adversarial Training via Model Calibration, Xiamen University, 2023.

Evaluation of Person Re-identification Robustness: Attack and Defense, ACCV 2022.

Deep Facial Synthesis: A New Challenge, the Chinese Academy of Science, 2022.

Adversarial Robustness in Computer Vision: Attack, Defense, and Beyond, ICIG 2021.

LANGUAGES

Chinese: Mother-tongue

English: Business

Japanese: Beginner